

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	Aswanth N N	Reg. No.:	192424123
Section / Batch:	2024	Date:	26-08-2026

Code of Conduct

I Aswanth N N (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Aswanth N N

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
----------------------	----------------------	---------------	--------

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

Answer:

Paragraph: "Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He (sentence 2)	Ravi, Arun
him (sentence 3)	Ravi, Arun
it (sentence 3)	book, library, Artificial Intelligence
they (sentence 4)	Ravi, Arun, {Ravi+Arun}
the book (sentence 4)	book on AI (already introduced)

b) Applying Constraints

- **Gender and number agreement:** Both Ravi and Arun are male singular, so "He/him" agree with both grammatically — this constraint alone cannot disambiguate. "They" must be plural, so it must refer to {Ravi, Arun} together — not to "it" or any singular entity.
- **Recency:** In sentence 2, the most recent NPs are "Ravi" and "Arun" (both introduced in sentence 1). Recency alone doesn't resolve it since both are equally close.
- **Grammatical role (Subject preference / Centering Theory):** Ravi is the subject of sentence 1 ("Ravi met Arun"), making Ravi the more likely discourse topic/focus. Subject-hood biases pronoun resolution toward Ravi for "He."
- **Semantic compatibility:** "It" must be compatible with an inanimate, findable object → "book" fits; "library" and "Artificial Intelligence" don't fit the verb "find it" as naturally (you find a book, not a library or a topic).
- **Discourse coherence:** Sentence 3 ("Arun helped him find it") makes sense only if "him" ≠ Arun (Arun can't help himself find something in this construction) → "him" = Ravi. This also reinforces "He" (sentence 2) = Ravi, since it's Ravi who was searching and Ravi who is being helped.

c) Resolution Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He (S2)	Ravi	Valid	Subject preference (grammatical role) + coherence with S3
He (S2)	Arun	Invalid	Arun is the one who "helps," not the one searching; violates coherence
him (S3)	Ravi	Valid	"Arun helped him" — Arun as helper implies the other person (Ravi) is helped

him (S3)	Arun	Invalid	Semantically odd for subject "Arun" to help himself in this frame
it (S3)	book on AI	Valid	Semantic compatibility — "find it" fits "book," an object of search
it (S3)	library / AI (topic)	Invalid	Libraries/topics aren't "found" this way; fails semantic compatibility
they (S4)	Ravi + Arun	Valid	Number agreement (plural); both are active discourse participants
they (S4)	book + AI	Invalid	These are inanimate and cannot "discuss" or "leave"

d) Final Coreference Chains

- **Chain 1 (Ravi):** Ravi → He (S2) → him (S3)
- **Chain 2 (Arun):** Arun → Arun (S3, repeated proper noun)
- **Chain 3 (book):** a book on Artificial Intelligence → it (S3) → the book (S4)
- **Chain 4 (Ravi + Arun):** they (S4)

e) Rewritten Paragraph (pronouns replaced)

"Ravi met Arun at the library. **Ravi** was looking for a book on Artificial Intelligence. Arun helped **Ravi** find **the book**. Later, **Ravi and Arun** discussed the book before leaving."

f) Ambiguity if "He was looking for a book" appeared *before* introducing Arun

If the sentence order were: "*He was looking for a book on Artificial Intelligence. Ravi met Arun at the library...*" — the pronoun "He" would have **no prior antecedent in the discourse** at all. This creates a **cataphoric/undefined reference** problem:

- The reader cannot resolve "He" to either Ravi or Arun because neither has been introduced yet.
- Discourse coherence breaks down since referring expressions typically require a prior (or at least contextually inferable) antecedent; without one, the reference is "dangling."
- This violates the general principle that pronouns are anaphoric (backward-looking) by default; using them cataphorically (forward-looking) is only acceptable in specific stylistic constructions (e.g., "Before he even entered the room, Ravi knew...") where the coreference is clarified almost immediately and unambiguously — which is not guaranteed here since two candidate names (Ravi, Arun) follow.

- Result: genuine referential ambiguity that a listener/reader (or an NLP coreference system) cannot resolve without additional cues.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

- 1. Maintain entity coherence using assignment, mistakes, and you.**
- 2. Use a Cause–Effect or Elaboration discourse relation.**
- 3. Include at least two keywords from:**
 - review
 - improve
 - confident
 - feedback
- 4. Response length must be 2–3 sentences.**
- 5. Maintain a positive and polite tone.**
- 6. Do not give contradictory advice.**

Tasks:

- a) Identify the dialog acts required.**
- b) Identify the important entities that must be maintained.**
- c) Generate three possible responses satisfying all constraints.**
- d) Compare the responses based on:**
 - Coherence
 - Constraint satisfaction

- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

Ans:

Input: Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

a) Dialog Acts Required

- **Reassure** — an emotional/affective act reducing the student's anxiety.
- **Advise** — a directive act suggesting a concrete next step.
- (Implicit) **Acknowledge** — recognizing the student's stated concern before reassuring, which supports coherence though not separately required.

b) Important Entities to Maintain

- **assignment** (the object submitted)
- **mistakes** (the source of worry)
- **you** (the student, addressee — maintains second-person continuity)

c) Three Candidate Responses

Response 1:

"Don't worry — submitting on time already shows good effort. If you're unsure about mistakes, you can review your answers and ask your instructor for feedback."

Response 2:

"It's completely normal to feel that way after submitting. A few mistakes won't undo your hard work, so try to stay confident and use any feedback you get to improve next time."

Response 3:

"I understand your concern, but worrying now won't change what's submitted. You should review the assignment guidelines carefully next time to feel more confident before submitting."

d) Comparison of Responses

Criterion	Response 1	Response 2	Response 3
Coherence (entity use: assignment, mistakes, you)	High — mentions "mistakes," "review," addresses "you" directly	High — mentions "mistakes," "you," "feedback," "improve"	Moderate — mentions "assignment" but shifts focus to "next time" rather than the current submitted assignment
Constraint satisfaction (2+ keywords: review, improve, confident, feedback)	Uses "review," "feedback" (2 keywords) ✓	Uses "confident," "feedback," "improve" (3 keywords) ✓	Uses "review," "confident" (2 keywords) ✓
Politeness	Warm, supportive	Warm, empathetic, validating	Slightly blunt ("worrying now won't change what's submitted") — mildly discouraging
Relevance (Cause–Effect/Elaboration)	Elaboration: cause (submitted) → effect (review/ask for feedback)	Elaboration: cause (feeling worried) → effect (stay confident, use feedback)	Cause-Effect present but advice is future-oriented, less relevant to <i>current</i> worry

e) Best Response and Justification

Best Response: Response 2

"It's completely normal to feel that way after submitting. A few mistakes won't undo your hard work, so try to stay confident and use any feedback you get to improve next time."

Justification:

- It satisfies **Reassure** ("completely normal to feel that way," "won't undo your hard work") and **Advise** ("use any feedback... to improve") clearly and in that order, matching a natural Reassure→Advise discourse flow.
- It uses **three** constraint keywords (confident, feedback, improve), exceeding the minimum requirement.

- It maintains **entity coherence** — "mistakes" and implicit "assignment" (via "hard work," "submitting") and "you" are all present.
- Tone is the most **empathetic and validating** of the three, avoiding the mildly discouraging framing in Response 3.
- It does not contradict itself — reassurance and advice align (unlike telling someone not to worry while also being blunt about the finality of submission).

f) What Happens if Entity Coherence Is Not Maintained

If entities like *assignment*, *mistakes*, and *you* are dropped or replaced inconsistently:

- The response may sound **generic and detached**, failing to feel like a direct reply to the student's specific concern (e.g., a reply that only says "Things will be fine!" without referencing the assignment or mistakes).
- It can cause **referential ambiguity** — the listener may not know what "it" or "that" refers to.
- **Discourse relations break down** (Cause-Effect/Elaboration require shared entities to link the worry to the advice); without shared entities, the advice may seem unrelated to the stated problem.
- Overall, the response loses **relevance and trustworthiness**, appearing as a template reply rather than a contextually grounded one — reducing the system's perceived understanding and empathy.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**

2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

- d) Generate a simple English paraphrase without ambiguity.
- e) Explain how Word Sense Disambiguation improves Machine Translation.

Ans:

Sentence: "Priya sat on the bank and watched the boats moving across the water."

a) Possible Meanings of "bank"

1. **Financial institution** — a place where money is deposited/withdrawn.
2. **Side/edge of a river** — the land bordering a body of water.

b) Correct Sense Selection and Justification

Selected sense: "side of a river"

Justification (contextual constraints):

- Co-occurring words "**boats**," "**moving across**," and "**water**" are strong semantic collocates of a riverine/water context, not a financial one.
- The verb "**sat on**" is physically compatible with a riverbank (a physical, sit-able surface) but incompatible with "a financial institution" as a literal physical object one sits on.
- **Selectional restrictions:** "watched the boats... across the water" requires a vantage point near a water body — consistent only with the river-side sense.
- No financial-domain words (money, account, loan, deposit) appear anywhere in the sentence, further ruling out the financial sense.

This is a classic case of disambiguation via **collocation and selectional preference** — the surrounding lexical items ("water," "boats") co-select the correct sense.

c) Predicate Logic Representation

Using the given predicates:

$\exists x \exists y \exists z \exists w [$

$\text{Priya}(x) \wedge$

$\text{bank}(y) \wedge$

$\text{sit}(x) \wedge \text{beside}(x, y) \wedge$

$\text{watch}(x, z) \wedge$

$\text{boat}(z) \wedge$

$\text{move}(z) \wedge$

$\text{water}(w) \wedge$

$\text{across}(z, w)$

$]$

Reading: "There exists an entity x (Priya), an entity y (a riverbank), an entity z (boats), and an entity w (water), such that Priya sits beside the bank, and Priya watches the boats, and the boats move across the water."

(Note: " $\text{sit}(x) \wedge \text{beside}(x,y)$ " is used instead of a single sit-on predicate since "beside" captures the riverbank sense — being alongside the water's edge — while preserving the given predicate inventory.)

d) Unambiguous English Paraphrase

"Priya sat by the riverside and watched the boats sailing across the water."

(Replacing "bank" with the unambiguous term "riverside" removes the lexical ambiguity while preserving meaning.)

e) How WSD Improves Machine Translation

- **Correct lexical choice in target language:** Many languages have entirely different words for different senses of a polysemous term (e.g., "bank" as riverbank vs. financial bank translates to distinct words in Hindi, French, German, etc.). WSD ensures the translator picks the semantically correct target word rather than a default/frequent one.
 - **Avoids mistranslation and loss of meaning:** Without WSD, an MT system might default to the most frequent sense (often "financial institution" for "bank"), producing a nonsensical or incorrect translation in a river context.
 - **Preserves contextual and discourse coherence:** Correct sense selection ensures that surrounding sentence elements (boats, water) still make sense together in the translated output, maintaining fluency and meaning across languages.
 - **Improves downstream NLP accuracy:** Since translation often relies on parsing, semantic role labeling, and predicate-argument structure (as shown in the logic representation above), correct sense resolution is a prerequisite for these deeper representations to be accurate, which in turn improves overall translation quality.
-

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	3	4	4	4	19	93
2	4	3	4	4	4	19	
3	4	4	4	4	3	19	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____